

Volume III.

Thirteenth Section. Equivalence according to number groups.**§ 98. Number groups in the orders.**

In order to investigate more closely the relations of quadratic irrational numbers to the orders of quadratic forms, we derive from the numbers of the field Ω number groups according to the following points of view.

1. We take an arbitrary natural number Q and exclude first, from the whole numbers in Ω , all those which are not prime to Q . From the remaining ones we also take arbitrary quotients.

The whole and fractional numbers η so obtained we call prime to Q . Let their totality be denoted by O . They form a group in so far as multiplication and division of two of these numbers always again give numbers of the same system.

2. We now proceed in the same way with the numbers of the order $[Q]$, i.e. we exclude all numbers of this order which are not relatively prime to Q , and also form the quotients of arbitrary two of the numbers left over. We denote these numbers by η' and their totality by O' , and call them whole and fractional numbers of the order $[Q]$. The numbers O' are all contained in O and likewise form a group with respect to multiplication and division. We call them the group of the order $[Q]$, or briefly an order group.

Every number η in O can be transformed into a whole number ω by multiplication with a whole rational number n prime to Q .

If η' is a number in O' , then one can rationalize the denominator and finds that $n\eta'$ is a whole number of the order $[Q]$ and hence is congruent to a rational number modulo Q .

Since n is prime to Q , one can put this number equal to nr , where r is a rational number prime to Q . Accordingly every number η' has the property

$$\eta' \equiv r \pmod{Q}, \quad (1)$$

where r is a whole rational number, and the congruence is to be understood in the sense $n\eta' = nr$.

The congruences (1) can be multiplied and divided, if by r^{-1} one understands the whole number satisfying the condition $rr^{-1} \equiv 1 \pmod{Q}$.

Conversely, a whole or fractional number η' which satisfies a condition (1) belongs to the group O' . For if $n\eta' = nr$ is a number in $[Q]$, then $\eta' = nr : n$ is a quotient of two such numbers.

3. In O' there is again a group O_1 , consisting of all numbers η_1 which satisfy the condition

$$\eta_1 \equiv 1 \pmod{Q}, \quad (2)$$

and the numbers of O_1 also give again numbers of O_1 under multiplication and division.

Thus we have three kinds of number groups, each of which is contained in the preceding one:

1. O contains all numbers in Ω prime to Q ,
2. O' contains the whole and fractional numbers of the order $[Q]$ prime to Q ,
3. O_1 contains the number classes in O' congruent to unity modulo Q .

The groups O, O', O_1 are infinite Abelian groups. But if one takes a complete residue system of rational numbers $r_1, r_2, \dots, r_\mu$ prime to Q , then every number in O' can be represented as a product of one of these r_i with a number in O_1 , and therefore one can put, in the terminology used in Vol. II, § 2,

$$O' = r_1 O_1 + r_2 O_1 + \dots + r_\mu O_1, \quad (3)$$

and μ is the index of the divisor O_1 of O' :

$$\mu = (O', O_1). \quad (4)$$

Thus μ is a finite number, namely

$$\mu = \varphi(Q) = Q \prod_q \left(1 - \frac{1}{q}\right), \quad (5)$$

where $\varphi(Q)$ is the usual symbol in number theory for the number of number classes modulo Q with numbers prime to Q , and where q runs in the product over all rational prime numbers occurring in Q .

If we denote by

$$\omega_1, \omega_2, \dots, \omega_\nu$$

a complete system of representatives of the numbers in Ω prime to Q , taken modulo Q , then in the same way

$$O = \omega_1 O_1 + \omega_2 O_1 + \dots + \omega_\nu O_1, \quad (6)$$

and

$$\nu = \psi(Q) = (O, O_1) \quad (7)$$

is the index of the divisor O_1 with respect to O .

The number $\psi(Q)$ was determined generally in Vol. II, § 168, (3), also for the case in which an ideal occurs in place of Q . Since here $N(Q) = Q^2$, we have

$$\nu = \psi(Q) = Q^2 \prod_{\mathfrak{q}} \left(1 - \frac{1}{N(\mathfrak{q})}\right), \quad (8)$$

where $\mathfrak{q}$ runs through the distinct ideal prime divisors of Q .

Now, by § 92,

$$\begin{aligned} 1. \quad N(\mathfrak{q}) &= q, & \text{if } \left(\frac{\Delta}{q}\right) &= +1, \\ 2. \quad N(\mathfrak{q}) &= q, & \text{if } \left(\frac{\Delta}{q}\right) &= 0, \\ 3. \quad N(\mathfrak{q}) &= q^2, & \text{if } \left(\frac{\Delta}{q}\right) &= -1. \end{aligned}$$

We shall denote these three kinds of prime numbers by q_1, q_2, q_3 , and remark that every prime number q_1 is the norm of two different prime ideals $\mathfrak{q}$. Accordingly from (8) one obtains

$$\psi(Q) = Q^2 \prod_{q_1} \left(1 - \frac{1}{q_1}\right)^2 \prod_{q_2} \left(1 - \frac{1}{q_2}\right) \prod_{q_3} \left(1 - \frac{1}{q_3}\right). \quad (9)$$

Now the index of O' with respect to O is, by Vol. II, § 2, (4),

$$(O, O') = \frac{(O, O_1)}{(O', O_1)} = \frac{\psi(Q)}{\varphi(Q)},$$

and

$$\frac{\psi(Q)}{\varphi(Q)} = Q \prod_{q_1} \left(1 - \frac{1}{q_1}\right) \prod_{q_3} \left(1 + \frac{1}{q_3}\right).$$

Therefore

$$(O, O') = Q \prod_{q_1} \left(1 - \frac{1}{q_1}\right) \prod_{q_3} \left(1 + \frac{1}{q_3}\right),$$

for which one can also put, according to the three cases,

$$(O, O') = Q \prod_{q|Q} \left(1 - \frac{1}{q} \left(\frac{\Delta}{q}\right)\right). \quad (10)$$

Here the product refers to all prime numbers q occurring in Q .

We can therefore choose a representative system

$$\eta_1, \eta_2, \dots, \eta_\lambda$$

in O such that

$$O = \eta_1 O' + \eta_2 O' + \dots + \eta_\lambda O' \quad (11)$$

where $\lambda = (O, O')$.

§ 99. Equivalence in the orders.

10. *Definition.* We shall now call two ideals $\mathfrak{a}, \mathfrak{a}'$ equivalent according to the order $[Q]$ if

$$\mathfrak{a}' = \eta' \mathfrak{a} \quad (1)$$

and η' denotes a number in O' . Here $\mathfrak{a}$ and $\mathfrak{a}'$ are always assumed to be prime to Q .

If one takes a number $\mu = \mathfrak{a}\mathfrak{m}$ divisible by $\mathfrak{a}$ in $[Q]$ in such a way that $\mathfrak{m}$ becomes a primary ideal, and puts $\mathfrak{a}' = \eta' \mathfrak{a}$, then

$$\mu' = \eta' \mu = \mathfrak{a}' \mathfrak{m} \quad (2)$$

and we can also explain the equivalence of $\mathfrak{a}$ and $\mathfrak{a}'$ according to $[Q]$ by saying that $\mathfrak{a}$ and $\mathfrak{a}'$ are transformed into numbers of the order $[Q]$ by multiplication with one and the same ideal $\mathfrak{m}$.

By the ideal $\mathfrak{m}$, according to § 97, a family of parallel quadratic forms (a, b, c) is determined, in which a and b are determined from

$$a = N(\mathfrak{m}), \quad \frac{b + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}. \quad (3)$$

And this family of forms therefore does not change by (2) if $\mathfrak{a}$ is replaced by an equivalent ideal $\mathfrak{a}'$.

If, however, in (2), one takes in place of the ideal $\mathfrak{m}$ another ideal $\mathfrak{m}_1$, and puts

$$\mu_1 = \mathfrak{a}\mathfrak{m}_1, \quad \mu'_1 = \mathfrak{a}'\mathfrak{m}_1, \quad (4)$$

then

$$\frac{\mu'_1}{\mu_1} = \frac{\mathfrak{a}'}{\mathfrak{a}} = \eta'$$

is a number in O' , and therefore $\mathfrak{m}_1$ is equivalent to $\mathfrak{m}$.

If then one puts

$$a_1 = N(\mathfrak{m}_1), \quad \frac{b_1 + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}_1}, \quad (5)$$

one obtains another primitive form (a_1, b_1, c_1) , of which we can prove that it is equivalent to (a, b, c) .

We put, for abbreviation,

$$A = \frac{b + \sqrt{D}}{2}, \quad A_1 = \frac{b_1 + \sqrt{D}}{2},$$

and remark that

$$N(A_1)A, \quad N(A)A_1$$

are numbers in $[Q]$ divisible by $\mathfrak{m}\mathfrak{m}_1$ [by (5)]. Accordingly, by § 97, 7., the whole rational numbers $\alpha, \beta, \gamma, \delta$ can be so determined that

$$A_1 = \alpha A + a(-\beta + \delta\theta), \quad (6)$$

and one proves quite as in § 95 that

$$\alpha\delta - \beta\gamma = 1$$

must hold, that is, the two forms

$$(a, b, c), \quad (a_1, b_1, c_1)$$

are equivalent. Thus we have:

11. *Theorem.* Every ideal class, represented by an ideal $\mathfrak{a}$ according to $[Q]$, corresponds to a form class A represented by the form (a, b, c) , and conversely.

In order to obtain the form class A , take a primary ideal $\mathfrak{m}$ of the reciprocal class A^{-1} and put

$$a = N(\mathfrak{m}), \quad \frac{b + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}.$$

Conversely, if the form (a, b, c) is given, then the greatest common divisor $\mathfrak{m}$ of

$$a, \quad \frac{b + \sqrt{D}}{2}$$

is an ideal of the class A^{-1} .